

Threats to Identification

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Part 4: Threats to Identification

1. Most Serious Threat: Selection into 401(k)-Offering Employers

- **Threat:** 401(k) eligibility is determined at the employer level, and workers are not randomly assigned to employers. Workers with stronger preferences for retirement savings may systematically seek out employers offering better benefits packages, including 401(k) plans. This creates a selection channel flowing through *unobservable* characteristics: even after conditioning on income, age, education, family size, marital status, IRA participation, and homeownership, workers with stronger savings preferences may be disproportionately concentrated in the eligible group. Our DML strategy assumes conditional independence, but this assumption is fundamentally untestable from survey data alone. A worker who has internalized strong savings norms from family background or financial literacy shocks will appear observationally identical to another worker yet differ dramatically in savings behavior. This residual unobserved heterogeneity causes our estimate to remain upward biased even after DML.
- **Direction of bias:** Upward. Workers who want to save more sort into 401(k)-offering firms and would save more regardless of plan access. Our estimate partially captures this preference channel, overstating the causal effect of eligibility per se.
- **What would address it:** The ideal strategy is an instrumental variable that shifts 401(k) availability independent of savings preferences. Abadie (2003) uses participation status (p401) as a fuzzy instrument for actual asset accumulation. A cleaner design would exploit a natural experiment: regulatory changes mandating 401(k) coverage for new categories of workers, or a regression discontinuity around employer eligibility thresholds tied to firm size or worker tenure.

2. Savings Crowd-Out Across Vehicles

- **Threat:** Even if 401(k) eligibility induces new saving behavior, some contributions may substitute for savings in other vehicles — IRAs, taxable brokerage accounts, money market funds. Our outcome `net_tfa` aggregates across financial asset categories to

capture the full portfolio effect. However, it is drawn from self-reported SIPP data, which systematically undercounts informal savings (cash, informal credit) and assets in non-standard accounts. If 401(k)-ineligible workers disproportionately hold underreported informal savings, the comparison group's assets are understated, inflating our estimate.

- **Why it matters:** If crowd-out is substantial and `net_tfa` misses substituted assets, we are measuring a reshuffling of existing savings, not new wealth creation.
 - **Partial mitigation:** Including IRA participation (`pira`) as a control partially accounts for the most common substitute. A more complete analysis would separately model 401(k), IRA, and taxable savings.
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3. Firm-Level Compensation Bundling

Our analysis treats 401(k) eligibility as an individual-level binary treatment, but it is a firm-level characteristic. Firms offering 401(k) plans may differ along multiple dimensions beyond income: job security, health insurance generosity, financial wellness programs, and employer match rates. Even after conditioning on income, workers at these firms may experience a richer compensation environment that independently increases savings. Direction of bias: upward — we cannot separate the 401(k) eligibility effect from the broader employer quality bundle.

4. What I Cannot Rule Out

Conditional independence — the foundation of our identification strategy — is untestable from the data we have. Our estimate should be interpreted as the ATE *under this assumption*, not a definitive causal claim. The stability across GBM (\$9,581) and RF (\$8,984) nuisance models validates the estimation procedure, not the identifying assumption. A bias sensitivity analysis shows the effect remains positive unless unobserved confounding accounts for more than ~75% of the DML estimate — an implausibly large residual given that our observables explain substantial variation in both treatment and outcome. External validity also warrants caution: the SIPP reflects a 1990s labor market. Today's environment — with widespread automatic enrollment and SECURE 2.0 provisions — may produce different behavioral responses to eligibility.